MUSTAFACODE

Lecture 01

Object-Orientation Programing (OOP)
Object-Oriented Programming (CS304)

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Lecture 1: Object-Orientation Programing (OOP)

Object-Orientation Programing (OOP)

Object-Orientation kya hai?

Yeh ek technique hai jisme hum apne programming problems ko **objects** ki shakal mein visualize karte hain aur unki **interactions** ko samajhte hain, bilkul real life ki tarah.

Examples

Hamari real life mein bohat se different objects hotay hain jo aik dusray ke sath interact karte hain taake different kaam perform kar sakein.

Misal ke taur par:

- Ali
- Car
- House
- Tree

Interactions:

- A person (Ali) ek house mein rehta hai
- A person car drive karta hai
- Tree bhi ek object hai

School ki Example

School mein bhi bohat se objects hotay hain jaise:

- Student
- Teacher
- Books
- Pen

- School Bag
- Classroom
- Parents
- Playground

Objects in a School

Agar hum school ke liye **fee collection system** banana chahte hain, to humein pehle:

- Related objects identify karne honge
- Unki interactions samajhni hongi

Bilkul real life ki tarah.

Conclusion

Is tarah hum keh sakte hain ke:

- Object-Oriented approach humein real world problems ko solve karna asaan banati hai
- Kyun ke hum problems ka solution **real world objects** ke terms mein sochte hain

Har cheez ko ek object samjha ja sakta hai:

- Har object ka apna behavior hota hai
- Har object ke kuch attributes hotay hain

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vs Procedural Paradigm

Object-Oriented:

- Focus on objects
- Objects ke beech interactions

Procedural:

- Focus on functions
- Functions ko main program mein call kiya jata hai

Model kya hai?

Model kisi real ya conceptual cheez ka **abstraction (simple representation)** hota hai.

Humein models ki zarurat hoti hai taake hum reality ke kisi hissa ko asaani se samajh sakein.

Model ki Examples:

- Highway maps
- Architectural models
- Mechanical models

OO Models (Object-Oriented Models)

Programming ke context mein models ka use problem ko samajhne ke liye kiya jata hai **development shuru karne se pehle**.

Hum Object-Oriented models banate hain jismein different **objects** aur unki **interactions** ko show kiya jata hai taake system ko samajhna asaan ho jaye.

Example 1 – Object Oriented Model

Objects:

- Ali
- Car
- House
- Tree

Interactions:

- Ali house mein rehta hai
- Ali car drive karta hai

Example 2 – Object Oriented Model (School Model)

Objects:

- Teacher
- Student
- School Bag
- Pen
- Book
- Playground

Interactions:

- Teacher student ko parhata hai
- Student ke paas school bag, book aur pen hota hai

Object-Orientation ke Advantages

Jaisa ke humne examples mein dekha, Object-Oriented models directly real life se relate karte hain, is liye:

- Hum asaani se kisi problem ka object-oriented model bana sakte hain
- Har koi is model ko asaani se samajh sakta hai
- Hum asaani se is model ko implement kar sakte hain kisi bhi object-oriented language (jaise C++) mein
  - Classes
  - Inheritance
  - Virtual Functions

Object kya hai?

Object ho sakta hai:

1. Tangible (Physical cheez)

- Ali
- School

- House
- Car

2. Conceptual (Sochi samjhi cheez)

- Time
- Date

Object ke Features

1. State (Attributes)

Object ki properties hoti hain (jaise color, name, size)

2. Well-defined Behavior (Operations)

Object kya kaam karta hai (jaise drive karna, parhna)

3. Unique Identity

Har object ki apni alag pehchaan hoti hai (chahe do objects same type ke kyun na ho)

Extra School Interaction

- Teacher → Student ko teach karta hai
- Student → School bag, book aur pen rakhta hai
- Student → Playground mein khelta hai

Tangible aur Intangible Objects

Tangible Objects ki Examples

Tangible objects wo hotay hain jo **physical form** mein mojood hotay hain (jinhein hum dekh ya touch kar sakte hain).

Example: Ali

Ali ek **tangible object** hai jiske kuch attributes aur behaviors hotay hain:

Characteristics (Attributes):

- Name
- Age

Behavior (Operations):

- Walk karta hai
- Khana khata hai

👉 Hum Ali ko uske **name** se identify karte hain.

Example: Car

Car bhi ek **tangible object** hai:

State (Attributes):

- Color
- Model

Behavior (Operations):

- Accelerate karna
- Car start karna
- Gear change karna

👉 Hum car ko uske **registration number** se identify karte hain.

Intangible Objects (Conceptual Objects) ki Examples

Yeh wo objects hotay hain jo **physical form mein nahi hotay**, sirf concept hote hain.

Example: Time

Time ek **intangible (conceptual) object** hai:

State (Attributes):

- Hours
- Minutes
- Seconds

Behavior (Operations):

- Set/Get Hours
- Set/Get Minutes
- Set/Get Seconds

👉 Time object ke liye hum apni taraf se **unique ID generate** karte hain.

Example: Date

State (Attributes):

- Year
- Month
- Day

Behavior (Operations):

- Set/Get Year
- Set/Get Month
- Set/Get Day

👉 Date object ke liye bhi hum apni taraf se **unique ID assign** karte hain.

Summary

- Model kisi real world scenario ka **abstraction** hota hai jo humein usay samajhne mein madad deta hai
- Object-Oriented model kisi problem ko **interacting objects** ki form mein describe karta hai
- Object Orientation ka use is liye hota hai kyun ke yeh real world problems ko programming mein map karna asaan banata hai
- Object Orientation **objects aur unke relationships** se achieve hoti hai
- Object ki properties uske **data members** se aur behavior uski **functions** se describe hota hai
- Objects **tangible (physical)** ya **intangible (conceptual/virtual)** ho sakte hain
- Kisi problem description mein jo **nouns** hotay hain, wo aksar objects ban sakte hain
- Ek object ke **multiple aspects** ho sakte hain
- Har object ka zaroori nahi ke system mein role ho (jaise school parking ka koi role na ho)

- Object-Oriented Programming se programs banana asaan hota hai kyun ke yeh **real life ke qareeb hota hai**